Privacy-Triggered Deferral in Legal E-Discovery: Risk-Aware Routing for PII Redaction

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Abstract

Automated PII redaction in legal e-discovery relies on NER models whose confidence scores proxy deployment safety. We show this proxy is insufficient: NER confidence clusters in a narrow high-confidence band (std $= 0.033$), providing almost no signal about document-level privacy risk. We propose **Privacy-Triggered Deferral**, a dual-objective Learning-to-Defer (L2D) policy that routes documents to human reviewers when either NER confidence is low *or* predicted privacy risk is high. Document-level sensitivity labels are generated by a local LLM (Qwen-2.5 7B); a multi-task DistilBERT (5) model is then trained with joint NER and risk-regression heads, distilling the LLM judgment into a deployable scorer. Under an e-discovery cost model (C_{leak}=\$50, C_h=\$5), our method reduces true leakage (sensitive documents auto-processed with incomplete redaction) from 5.5% to 0.4% and sensitive-document exposure from 22.7% to 4.6% at τ_c=0.85 (mean over 5 seeds) at comparable cost (\$4.48 vs. \$4.57/doc). At tighter thresholds, true leakage reaches zero. These results suggest that routing policies should incorporate risk signals beyond model confidence in high-stakes domains.

1 Introduction

In legal e-discovery, organizations must redact Personally Identifiable Information (PII) from large document corpora before production to opposing counsel. Manual review costs \$5–\$50 per document (8), motivating AI-based NER pipelines. Standard L2D and Selective Prediction frameworks (1; 2) route documents to human reviewers only when model confidence is *low*, assuming high confidence implies safe automation.

This assumption has a critical blind spot. A model may be highly confident yet the document may contain high-severity PII (SSNs, credit card numbers) or multiple co-occurring identifier types that amplify re-identification risk if any token is missed—a *collateral privacy leak*. Because PII entities have clear orthographic signatures, NER confidence clusters in a narrow band (std $= 0.033$), providing almost no discriminative signal between high-risk and low-risk documents.

Effective human-AI collaboration requires more than routing on uncertainty alone (4). In practice, deployed AI systems use context-dependent guardrails that go beyond confidence thresholds (3); our work operationalises this insight into a mathematically precise routing policy. We propose **Privacy-Triggered Deferral**, a dual-objective L2D policy with a second trigger based on predicted contextual privacy risk, directly implementing the *Conditionally Autonomous AI* oversight level (3) where capability does not imply permission.

1.1 Related work

Standard L2D optimises a joint loss over model and expert predictions (2); selective prediction abstains when confidence is low (1); both assume confidence is the sole routing signal. Our work introduces a *second*, context-aware routing axis orthogonal to confidence and specific to the privacy domain.

1.2 Contributions

(i) We identify *confidence insufficiency*: the narrow NER confidence band renders confidence-only deferral a poor discriminator of document-level privacy risk. **(ii)** We propose a dual-objective routing policy $\text{Defer}(x) = \mathbf{1}[c(x) < \tau_c \ \vee \ r(x) > \tau_r]$ that reduces true leakage by $\sim 14\times$ relative to confidence-only deferral. **(iii)** We train a multi-task DistilBERT (5) with joint NER and risk-regression heads, where sensitivity labels are distilled from a local LLM. **(iv)** We show this leakage reduction comes at comparable cost, making it practical for privacy-conscious deployment.

2 Method

2.1 Dataset

We use the `ai4privacy/pii-masking-400k` dataset (6), selecting 500 English-language synthetic documents with character-level PII annotations across 17 entity types. Documents are split 64%/16%/20% per seed; all results are mean $\pm$ std over 5 seeds ($\sim$47 of 100 test documents are sensitive).

2.2 Sensitivity labels

The dataset provides PII spans but no document-level sensitivity scores. For each document, a Qwen-2.5 7B model (7) receives the full text and PII types and rates contextual sensitivity from 0 to 1, considering PII severity, surrounding topic sensitivity, re-identification risk, and PII type diversity. Documents with $s(x) \geq 0.5$ are labeled sensitive (240/500, 48%). The LLM is used only during data preparation; at inference time, only the DistilBERT risk head is used (see Appendix A for the score distribution). Figure 1 illustrates the two-stage pipeline.

2.3 Model architecture

A shared `distilbert-base-uncased` backbone (5) (66M parameters) feeds two heads: a **NER head** producing per-token redact/keep logits with confidence $c(x)$ as the mean max-probability over entity-predicted tokens, and a **Risk head** operating on `[CLS]` to produce $r(x) \in [0, 1]$. Both are trained jointly: $\mathcal{L} = \mathcal{L}_{\text{NER}} + \mathcal{L}_{\text{Risk}}$ (cross-entropy + MSE).

2.3.1 Cost model

Let $c(x)$ be NER confidence and $r(x)$ the predicted risk. The human reviewer is modeled as an oracle (2). Per-document cost is:

$$\text{Cost}(x) = \begin{cases} C_h & \text{if } d(x) = \text{defer} \\ \underbrace{\sum_t \mathbf{1}[\hat{y}_t \neq y_t] \cdot C_{\text{err}}}_{\text{missed-token cost}} + \underbrace{\mathbf{1}[\text{sens.}] \cdot \mathbf{1}[\text{miss} > 0] \cdot r(x) \cdot C_{\text{leak}}}_{\text{leakage cost (if redaction incomplete)}} & \text{if } d(x) = \text{automate} \end{cases} \tag{1}$$

with C_h=\$5, C_{err}=\$2, C_{leak}=\$50 (8).

2.3.2 Routing policies

We compare three strategies of increasing oversight:

(1) Autonomous AI : Automate all documents; $\text{Defer}(x) = \text{False} \ \forall x$.

(2) Confidence-based L2D : Defer when confidence is low (standard selective prediction):

$$\text{Defer}(x) = \mathbf{1}[c(x) < \tau_c] \tag{2}$$

(3) Privacy-Triggered L2D (proposed) **:** Defer when confidence is low *or* risk is high:

$$\boxed{\text{Defer}(x) = \mathbf{1}[c(x) < \tau_c \ \vee \ r(x) > \tau_r]} \tag{3}$$

The OR condition routes high-confidence documents that are contextually dangerous to a human reviewer—a capability Eq. 2 structurally cannot provide.

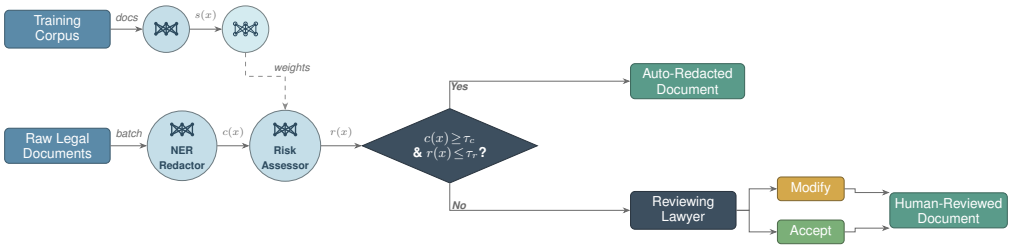

Figure 1: Privacy-Triggered Deferral: two-stage pipeline. Diagram elements follow the Fabric AI workflow legend (3) (Input/Output blocks, fixed vs. training model circles with network icon, governance mechanism, decision diamond, Accept/Modify actions, final outcomes). *Top:* During training, a local LLM annotates documents with sensitivity scores $s(x)$; the DistilBERT risk head regresses these. *Bottom:* At deployment, only DistilBERT is used. The decision point routes documents autonomously only if both safety checks pass.

Table 1: Policy comparison at representative thresholds (C_h=\$5), mean $\pm$ std over 5 seeds. *Exposure*: % of sensitive docs auto-processed. *Leakage*: % with incomplete redaction.

Policy	Auto. (%)	Expos. (%)	Leak. (%)	F1	Cost (\$/doc)
Autonomous	100	100	35.7±8.2	.915±.012	5.53±0.92
Conf. (τ_c=0.85)	25.6±6.5	22.7±5.7	5.5±3.9	.986±.009	4.57±0.39
Conf. (τ_c=0.90)	2.8±2.2	3.2±1.4	0.4±0.8	.999±.002	4.92±0.07
Priv. (τ_c=0.85, τ_r=0.5)	**13.2±6.5**	**4.6±4.3**	**0.4±0.9**	**.992±.007**	**4.48±0.39**
Priv. (τ_c=0.75, τ_r=0.4)	6.8±4.7	0.8±1.1	0.0±0.0	.998±.003	4.68±0.21
Priv. (τ_c=0.90, τ_r=0.5)	1.2±1.6	0.4±0.8	0.0±0.0	.999±.001	4.95±0.07

3 Experiments

3.1 Setup

We sweep $\tau_c \in \{0.75, \ldots, 0.95\}$, $\tau_r \in \{0.2, \ldots, 0.6\}$, and $C_h \in \{\$2, \ldots, \$40\}$ (full grids in Appendix F). Key metrics: *automation rate*, *sensitive-document exposure* (% sensitive docs auto-processed), *true leakage rate* (% sensitive docs auto-processed *and* incompletely redacted), *system F1*, and *expected cost* (Eq. 1). On a representative seed, NER confidence spans $[0.725, 0.900]$ (mean $= 0.826$, std $= 0.033$); sensitive and non-sensitive documents have nearly identical mean confidence (0.825 vs. 0.827), confirming confidence alone cannot separate them.

3.2 Results

Table 1 reports results at representative thresholds. At τ_c=0.85, confidence-only exposes 22.7% of sensitive documents, of which 5.5% suffer incomplete redaction (true leakage). Privacy-triggered at the same τ_c with τ_r=0.5 cuts exposure to 4.6% and true leakage to 0.4%—a $\sim 14\times$ reduction—at comparable cost (\$4.48 vs. \$4.57/doc). At tighter thresholds (τ_c=0.75, τ_r=0.4), true leakage reaches 0.0%.

Figure 2 shows the core insight: all documents cluster in a narrow high-confidence band, making a vertical confidence boundary ineffective. The L-shaped privacy-triggered boundary captures high-risk documents in the upper-right quadrant and routes them to human review. The primary benefit is *true leakage reduction* rather than cost savings; the cost advantage is modest and seed-dependent (see Appendix C for cost decomposition and Appendix D for Pareto frontiers).

4 Discussion and Conclusion

The narrow NER confidence distribution (std $= 0.033$) is a structural property of well-annotated PII data, where entity boundaries are orthographically unambiguous. Confidence-only deferral cannot distinguish sensitive from non-sensitive documents (mean confidence: 0.825 vs. 0.827); the risk head captures what confidence cannot. Privacy-Triggered Deferral benefits documents in the

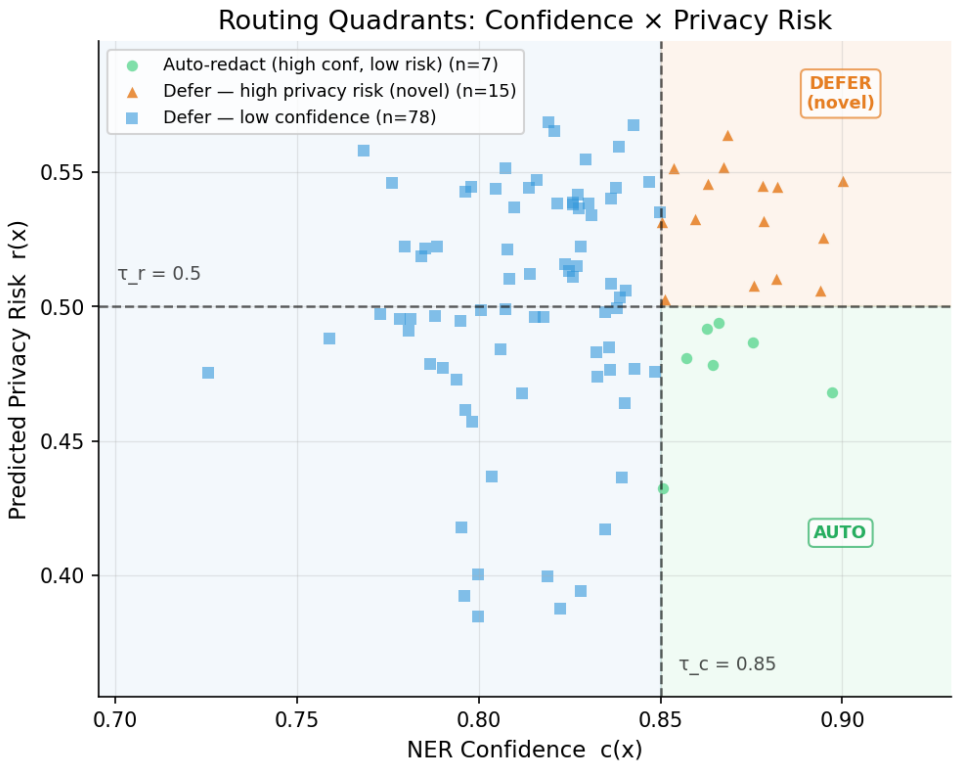

Figure 2: Routing quadrants: confidence × predicted risk (seed 42, 100 test documents). Documents cluster in a narrow high-confidence band ($c(x)>0.72$); a vertical confidence threshold cannot separate risk levels. The L-shaped privacy-triggered boundary routes the upper-right quadrant (high risk) to human review.

high-confidence, high-risk region—e.g., documents containing SSNs alongside other identifiers, where entity detection is trivial but combined context amplifies exposure if any token is missed (see Appendix E for detailed analysis including failure cases).

The fundamental trade-off is *true leakage vs. automation rate* under asymmetric penalties. The $\tau_r{=}0.4$ operating point provides a conservative zero-leakage default; raising to $\tau_r{=}0.5$ increases automation to 13% but introduces 0.4% true leakage on average (Appendix F.3). The autonomous baseline risks *automation bias*; confidence-only at $\tau_c{=}0.95$ exhibits *under-reliance*; Privacy-Triggered Deferral targets a middle ground.

4.1 Limitations

Sensitivity labels are generated by a 7B-parameter LLM, not human experts; low true leakage reflects successful approximation of LLM judgment, not elimination of contextual privacy risk. The evaluation uses 500 synthetic documents with 100-document test sets; validation on real legal corpora is needed. The oracle human assumption (100% accuracy) overestimates real reviewer performance; a noisy-human model would shift optimal thresholds.

4.2 Conclusion

We demonstrated that model confidence is an insufficient routing proxy in legal e-discovery. Privacy-Triggered Deferral reduces true leakage by $\sim 14\times$ relative to confidence-only deferral (0.4% vs. 5.5%, mean over 5 seeds) at comparable cost, while cutting sensitive-document exposure from 22.7% to 4.6%. The core insight is that *capability does not imply permission*: routing policy design should encode risk signals beyond model confidence in high-stakes domains.

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A Dataset Distribution

Figure 3 visualises the LLM-annotated sensitivity score distribution and PII type frequencies. Of 500 documents, 240 (48%) are labeled sensitive ($s(x) \geq 0.5$) and 260 (52%) non-sensitive. High-risk PII types (SOCIALNUM, CREDITCARDNUMBER, IDCARDNUM) contribute to higher sensitivity scores despite lower raw frequency than GIVENNAME/SURNAME. Because NER confidence is blind to PII *type*—a name and an SSN look equally unambiguous to the NER head—the confidence axis alone cannot distinguish the two clusters.

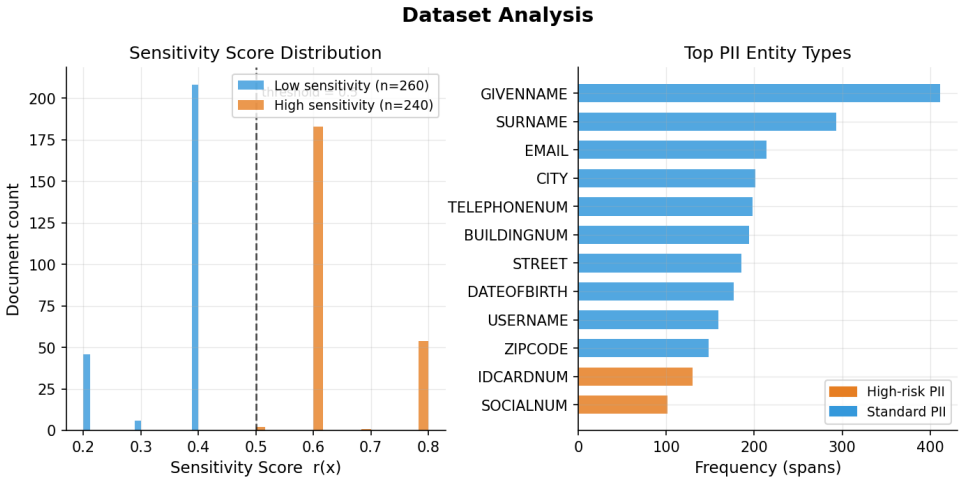

Figure 3: Dataset: LLM-annotated sensitivity distribution and PII type frequencies. *Left:* Sensitivity scores from Qwen-2.5 7B; 240 documents (48%) are labeled sensitive ($s(x) \geq 0.5$). *Right:* High-risk types (orange) contribute to higher LLM sensitivity scores despite lower raw frequency.

B Policy Comparison Visualisation

Figure 4 visualises the policy comparison at matched thresholds (τ_c=0.85, C_h=\$5).

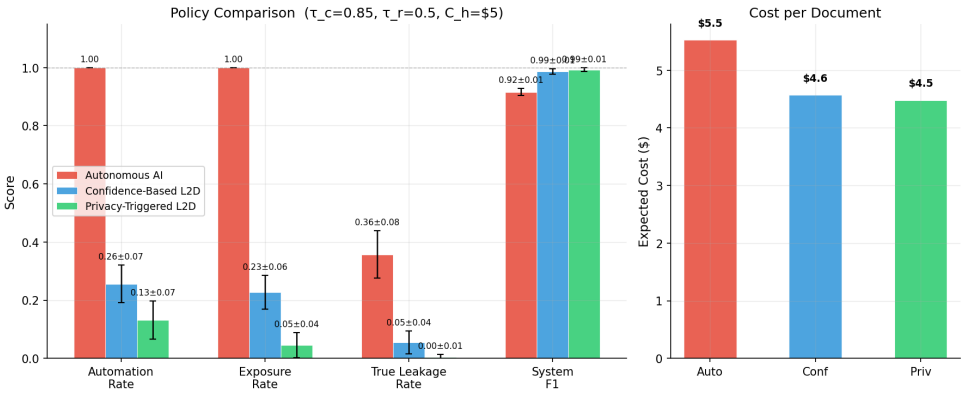

Figure 4: Policy comparison (mean $\pm$ std over 5 seeds, τ_c=0.85, C_h=\$5). Privacy-triggered (τ_r=0.5) cuts sensitive-document exposure from 22.7% to 4.6% and true leakage from 5.5% to 0.4% at comparable cost. Error bars show standard deviation across seeds.

C Cost Decomposition

Table 2 decomposes per-document cost. The autonomous baseline pays a large leakage cost (\$4.47/doc). Confidence-only and privacy-triggered achieve similar totals, but privacy-triggered substitutes human-review cost for leakage risk, nearly eliminating the leakage component. Figure 5 visualises this decomposition.

Table 2: Cost decomposition at τ_c=0.85, C_h=\$5 (mean over 5 seeds). Leakage cost is conditional on incomplete redaction.

Policy	Review	Miss	Leak	Total
Autonomous	\$0.00	\$1.06	\$4.47	\$5.53
Conf. (τ_c=0.85)	\$3.72	\$0.16	\$0.69	\$4.57
Priv. (τ_c=0.85, τ_r=0.5)	\$4.34	\$0.10	\$0.05	\$4.48

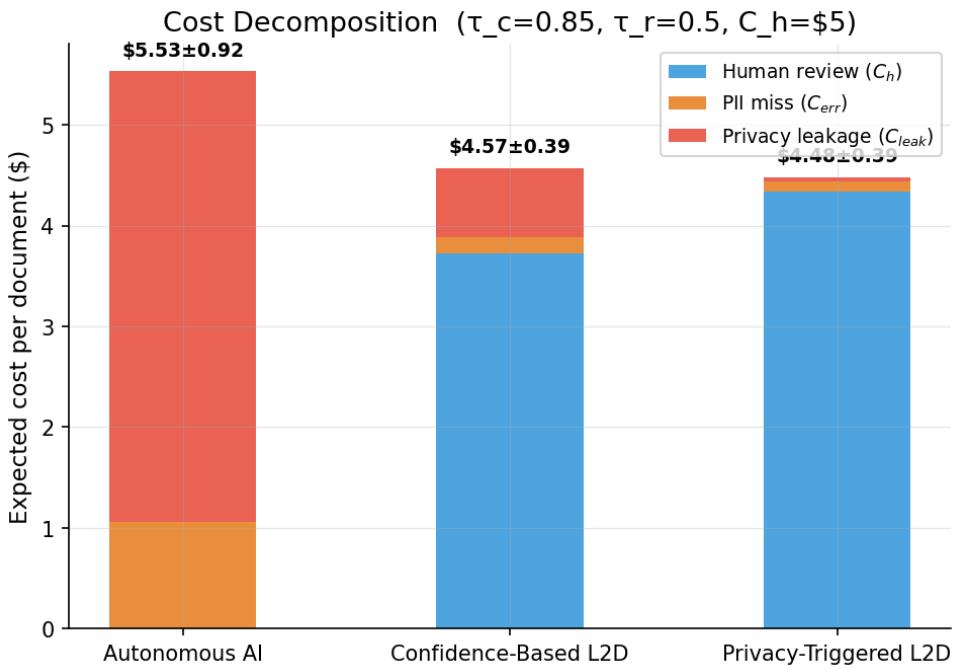

Figure 5: Cost decomposition by policy (mean over 5 seeds). Leakage penalties account for $\sim$80% of autonomous cost. Privacy-Triggered Deferral nearly eliminates leakage cost; its total cost is comparable, dominated by human review.

D Pareto Frontiers

In the automation–leakage plane (Fig. 6), the privacy-triggered curve generally lies below the confidence-only curve: at comparable automation rates it achieves substantially lower true leakage. In the cost–F1 plane (Fig. 7), both L2D strategies achieve F1$\geq$0.98 at costs near \$5.00/doc, while the autonomous baseline sits at F1$\approx$0.915 and \$5.53/doc.

E Detailed Analysis

Why confidence alone fails. The confidence distribution (std $= 0.033$, min $= 0.725$) means all documents appear safe by confidence alone—a structural property of well-annotated PII data where entity boundaries are orthographically unambiguous. The confidence-only policy draws a vertical boundary in $(c(x), r(x))$ space (Fig. 2), but sensitive and non-sensitive documents are distributed nearly uniformly across the confidence range (mean: 0.825 vs. 0.827), making them indistinguishable. The risk head captures what confidence cannot: an estimate of document-level privacy severity.

When the risk trigger helps most. Privacy-Triggered Deferral benefits documents in the high-confidence, high-risk region—e.g., a document containing an SSN alongside other identifier types, where entity detection is trivial but combined context amplifies exposure if any token is missed.

Failure cases and variance. With 320 training examples, the risk head's discrimination (gap $= 0.047$) is modest, and results show meaningful variance (e.g., exposure $4.6\%\pm4.3\%$, true leakage

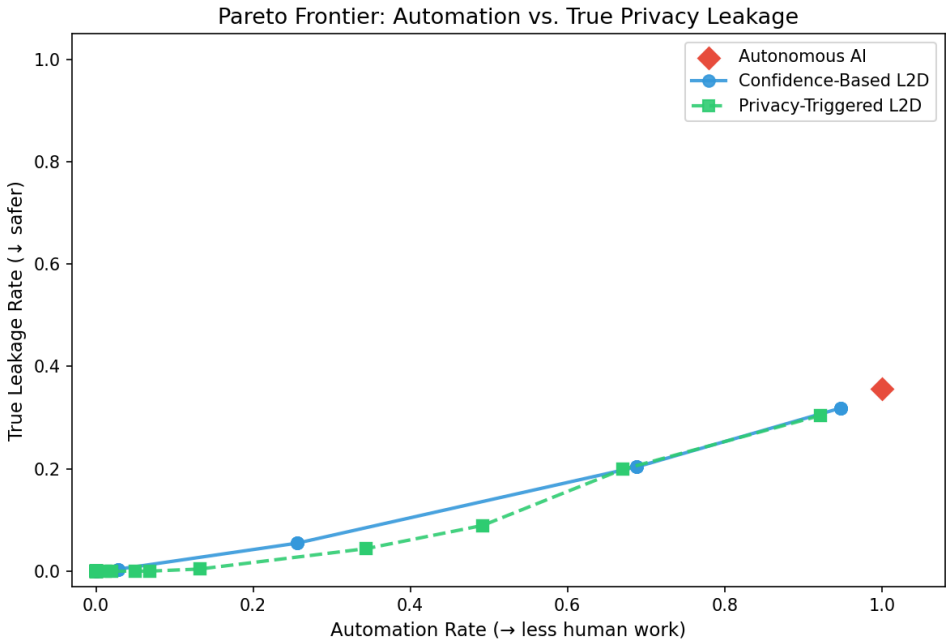

Figure 6: Pareto frontier: automation rate vs. true leakage rate (mean over 5 seeds). Privacy-triggered (green) generally achieves lower true leakage than confidence-only (blue) at comparable automation rates.

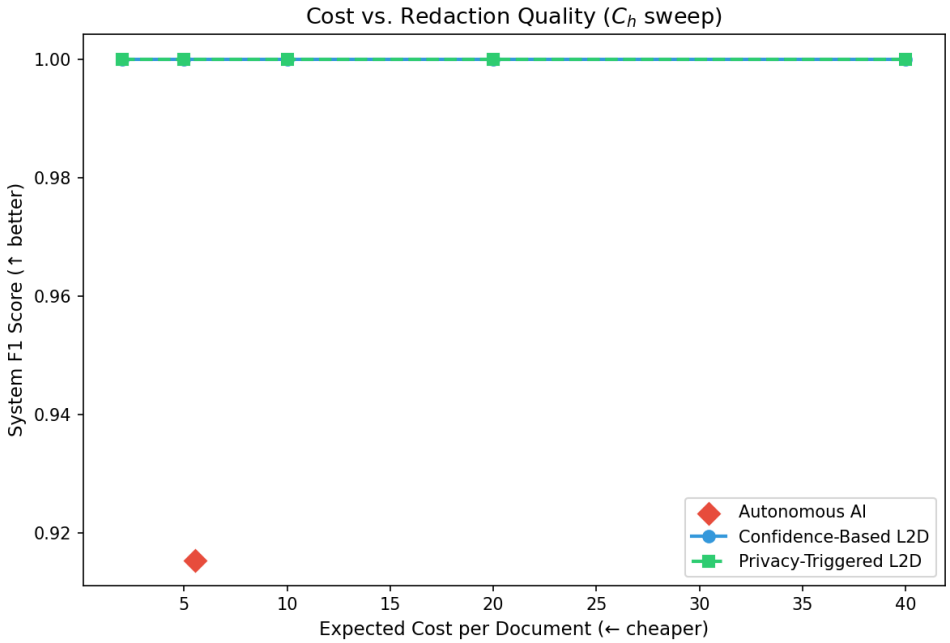

Figure 7: Pareto frontier: expected cost vs. system F1 (mean over 5 seeds). Both L2D strategies achieve F1≥ 0.98; the autonomous baseline sits at F1≈ 0.915 and $\$5.53$/doc.

$0.4\% \pm 0.9\%$ at τ_c=0.85, τ_r=0.5). On most seeds the risk trigger achieves zero true leakage; on occasional seeds a single sensitive document slips through. The model tends toward over-flagging rather than under-flagging—the correct failure mode for a safety-critical system.

Routing trade-offs: over- and under-reliance. The autonomous baseline risks *automation bias*: legal teams may assume the AI handles sensitivity and omit audits. Confidence-only at τ_c=0.95 exhibits *under-reliance*: sending every document to human review defeats the system's purpose. Privacy-Triggered Deferral targets a middle ground. The oracle human (100% accuracy) overestimates real reviewer performance; future work should replace it with a noisy-human model.

Transparency and calibration. The routing decision (Eq. 3) is interpretable: a deferred document carries an explanation—either model uncertainty or high sensitivity classification. Future work should apply temperature scaling to $r(x)$ so that τ_r maps to an explicit false-negative rate guarantee.

F Threshold Sweeps

F.1 Confidence-Based L2D Full Sweep

Table 3: Confidence-only threshold sweep (C_h=\$5, mean over 5 seeds). Expos. = sensitive docs automated; Leak. = sensitive docs automated **and** incompletely redacted.

τ_c	Auto.	Expos.	Leak.	F1	Cost
0.75	95%	92.2%	31.9%	0.924	\$5.19
0.80	69%	65.3%	20.4%	0.951	\$4.73
0.85	26%	22.7%	5.5%	0.986	\$4.57
0.90	3%	3.2%	0.4%	0.999	\$4.92
0.95	0%	0.0%	0.0%	1.000	\$5.00

F.2 Privacy-Triggered L2D Threshold Grid

Table 4: Selected operating points, privacy-triggered L2D (C_h=\$5, mean over 5 seeds).

τ_c	τ_r	Auto.	Expos.	Leak.	F1	Cost
0.90	0.5	1%	0.4%	0.0%	0.999	\$4.95
0.85	0.5	13%	4.6%	0.4%	0.992	\$4.48
0.75	0.5	49%	24.1%	8.9%	0.959	\$4.03
0.80	0.6	67%	61.2%	19.9%	0.952	\$4.76
0.75	0.6	92%	86.3%	30.4%	0.925	\$5.12

F.3 Risk Threshold Sensitivity

Table 5 isolates the sensitivity of τ_r at fixed τ_c=0.85. Figure 8 sweeps τ_r, showing τ_r=0.4 as the cost-optimal zero-leakage threshold (\$4.91/doc, 2% automation, 0.0% true leakage). Raising τ_r to 0.5 increases automation to 13% but introduces 4.6% exposure and 0.4% true leakage on average; further raising to 0.6 brings 21.4% exposure, 5.5% true leakage, and *increases* total cost to \$4.60 due to leakage penalties. This non-monotonic cost curve illustrates why naive automation-rate maximisation is the wrong objective in privacy-critical settings.

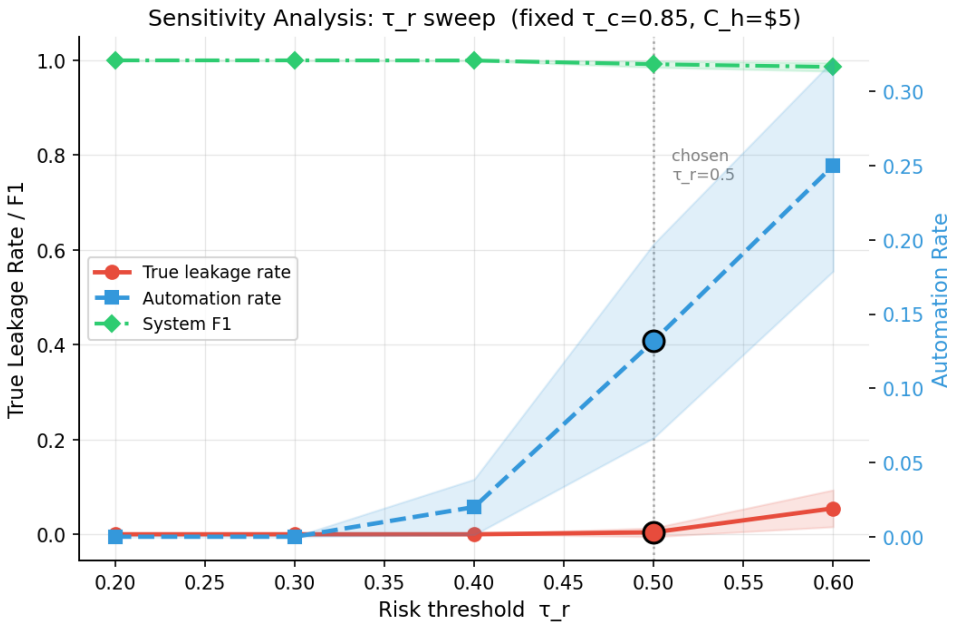

Figure 8: Sensitivity analysis: sweeping τ_r at fixed τ_c=0.85 (mean over 5 seeds; shaded bands show $\pm$1 std). τ_r=0.4 is the cost-optimal zero-true-leakage point. Raising τ_r to 0.6 increases automation but raises cost due to leakage-cost penalties, demonstrating the non-monotonic cost–automation relationship.

Table 5: τ_r sweep at fixed τ_c=0.85, C_h=$5 (mean over 5 seeds).

τ_r	Auto.	Expos.	Leak.	F1	Cost
0.2	0%	0.0%	0.0%	1.000	$5.00
0.3	0%	0.0%	0.0%	1.000	$5.00
0.4	2%	0.0%	0.0%	1.000	$4.91
0.5	13%	4.6%	0.4%	0.992	$4.48
0.6	25%	21.4%	5.5%	0.986	$4.60